

Date types(variable type)

The basic distinction is between

quantitative variables (for which one asks “how much?”) and

categorical variables (for which one asks “what type?”).

(1) Quantitative variables

(a)Discrete number- counted as whole no.

exp:- No. of kids 2, 3, Training session 1,2,3 its always integer not a flaot.(2.5)

(b)Continuous:- Number can be infinite precision .

Exp:-you can sleep 7 hr 30 min & 20 sec, distance can be 70.97 meters

(2) Categorical variables

Can not be measure against each other like color (blue, red, yellow)

(4)Ordinal

-Result rank A++,A+,A,B+.. here it can be compare among them.

Examples of types of data	
Quantitative	
Continuous	Discrete
Blood pressure, height, weight, age	Number of children Number of attacks of asthma per week
Categorical	
Ordinal (Ordered categories)	Nominal (Unordered categories)
Grade of breast cancer Better, same, worse Disagree, neutral, agree	Sex (male/female) Alive or dead Blood group O, A, B, AB

Variables shown at the left of Table can be converted to ones further to the right by using “cut off points”.

For example, blood pressure can be turned into a nominal variable by defining “hypertension” as a diastolic blood pressure greater than 90 mmHg, and “normo-tension” as blood pressure less than or equal to 90 mmHg.

Height (continuous) can be converted into “short”, average” or “tall” (ordinal).

Basic measurement of Statistics

Mean, median, Mode, Variance, Standard Deviation, Quartile Range...

df.describe():-function to summarize information

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In [55]: df1.describe()
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Out[55]:

	Profit	Unit Price	Shipping Cost
count	17.000000	18.000000	17.000000
mean	305.741382	75.768333	17.985294
std	884.219143	119.998788	23.028008
min	-481.041000	2.880000	0.700000
25%	-11.682000	4.987500	2.990000
50%	26.920000	39.415000	5.260000
75%	313.578000	99.722500	26.220000
max	3424.220000	500.980000	74.350000

- **Counts :-No. of observations**
- **Average value (mean)**
- **Std:-On average, how much each measurement deviates from the mean (standard deviation of the mean)**
- **Min:- minimum value of the observation.**
- **Max:- Maximum value of the observation.**
- **25,50,75 percentiles. (Quartile Range)**
- **Most frequently occurring value (mode)**
- **Midpoint between the lowest and highest value of the set (median)**

Mean :-

Mean is the average of the given data set.

Mean for ungrouped data = Sum of observations/Number of observations

1. Find the mean of the first 10 odd integers.

Solution:

First 10 odd integers: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19

Mean = Sum of the first 10 odd integers/Number of such integers

$$= (1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19)/10$$

$$= 100/10$$

$$= 10$$

Therefore, the mean of the first 10 odd integers is 10.

Median:-

The median is the middlemost data value of an ordered data set. This can be estimated using the formulas given below.

When n is odd:

$$\text{Median} = (n + 1)/2\text{th observation}$$

When n is even:

$$\text{Median} = [(n/2)\text{th observation} + \{(n/2)+1\}\text{th observation}]/2$$

2. What is the median of the following data set?

32, 6, 21, 10, 8, 11, 12, 36, 17, 16, 15, 18, 40, 24, 21, 23, 24, 24, 29, 16,
32, 31, 10, 30, 35, 32, 18, 39, 12, 20

Solution:

The ascending order of the given data set is:

6, 8, 10, 10, 11, 12, 12, 15, 16, 16, 17, 18, 18, 20, 21, 21, 23, 24, 24, 24, 29, 30, 31, 32, 32, 32, 35, 36, 39, 40

Number of values in the data set = $n = 30$

$$n/2 = 30/2 = 15$$

15th data value = 21

$$(n/2) + 1 = 16$$

16th data value = 21

Median = $[(n/2)\text{th observation} + \{(n/2)+1\}\text{th observation}]/2$

$$= (15\text{th data value} + 16\text{th data value})/2$$

$$= (21 + 21)/2$$

$$= 21$$

Mode:-

Mode is the most frequently occurring value in the data set.

3. Identify the mode for the following data set:

21, 19, 62, 21, 66, 28, 66, 48, 79, 59, 28, 62, 63, 63, 48, 66, 59, 66, 94, 79, 19, 94

Solution:

Let us write the given data set in ascending order as follows:

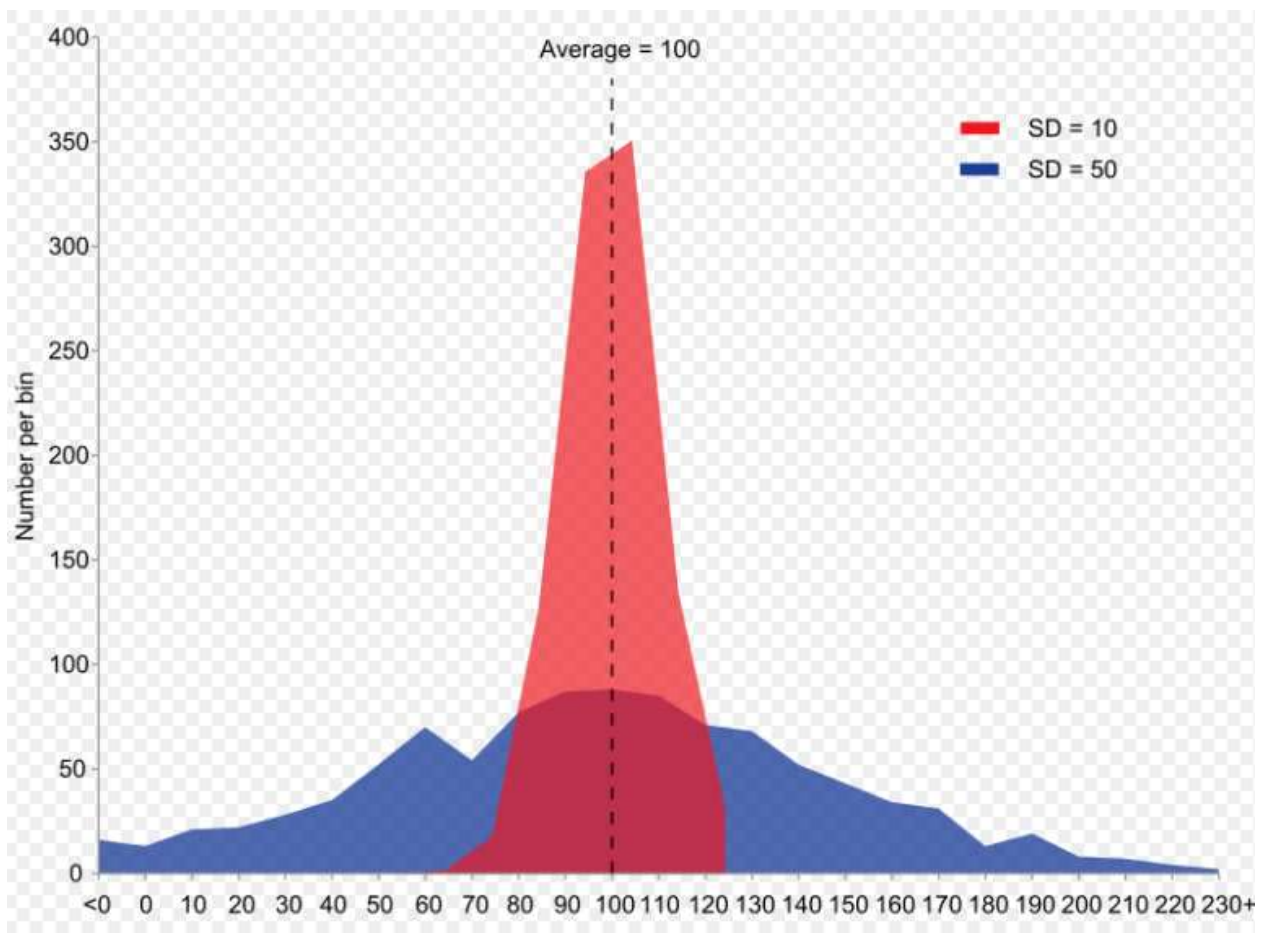
19, 19, 21, 21, 28, 28, 48, 48, 59, 59, 62, 62, 63, 63, 66, 66, 66, 66, 79, 79, 94, 94

Here, we can observe that the number 66 occurred the maximum number of times. Thus, the mode of the given data set is 66.

Variance

Dispersion

Dispersion (also called variability, scatter, or spread) is the extent to which a distribution is stretched or squeezed. Common examples of measures of statistical dispersion are the variance, standard deviation, and interquartile range. For instance, when the **variance** of data in a set is **large**, the data is **widely scattered**. On the other hand, when the **variance** is **small**, the data is less scattered.



Example:-

Cricketer IPL Match

Component life of an equipment

Pizza Delivery

Variance of Population

$$\sigma^2 = \frac{\sum (x_i - \mu)^2}{N}$$

σ^2 = Population Variance
 N = the size of the population
 x_i = each value from the population
 μ = the population mean

Standard Deviation: -

It is the square root of variance.

$$\sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

σ = population standard deviation
 N = the size of the population
 x_i = each value from the population
 μ = the population mean

standard deviation is a measure of the amount of variation or dispersion of a set of values. A low standard deviation indicates that the values tend to be close to the mean (also called the expected value) of the set, while a high standard deviation indicates that the values are spread out over a wider range.